

Electromagnetism 2

(spring semester 2026)

Lecture 8

Electric field in dielectric materials

- ❖ The electric displacement field $\mathbf{D}$; relative permittivity
- ❖ Gauss law for the $\mathbf{D}$ field
- ❖ Boundary conditions for the $\mathbf{E}$ and $\mathbf{D}$ fields
- ❖ Energy density of the electric field

Previous lecture

- ❖ Polarisation of LHM dielectric materials, characterised by the electric susceptibility χ_E :

$$\vec{P} = \frac{1}{\Delta V} \sum_{\Delta V} \vec{p}_i = \chi_E \cdot \epsilon_0 \cdot \vec{E}$$

- ❖ Volume distribution of polarisation $\mathbf{P}$ is equivalent to the following distribution of surface and volume charges:

$$\sigma_p = \vec{P} \cdot \vec{n} \quad \text{and} \quad \rho_p = -\nabla \cdot \vec{P}$$

- ❖ Electric susceptibility for a dielectric gas:

$$\chi_E = n \left(\alpha + \frac{p^2}{3\epsilon_0 kT} \right)$$

The two terms are due to *electronic polarisation* and the *alignment of permanent dipole moments p* of molecules.

The electric displacement field

Dielectric materials in non-uniform fields: (lecture 7)
induced *volume polarisation charges* with density $\rho_p = -\nabla \vec{P}$

In general, dielectrics also carry *free charges* with a density ρ_f

Total charge density: $\rho = \rho_f + \rho_p$

Gauss law: $\nabla \vec{E} = \frac{\rho}{\epsilon_0} = \frac{1}{\epsilon_0} (\rho_f + \rho_p) = \frac{1}{\epsilon_0} (\rho_f - \nabla \vec{P})$

Therefore $\epsilon_0 \nabla \vec{E} + \nabla \vec{P} = \rho_f$ and $\nabla (\epsilon_0 \vec{E} + \vec{P}) = \rho_f$

Let's define the *electric displacement field*:

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

Gauss law for the D field

Gauss law for the electric displacement field: $\nabla \vec{D} = \rho_f$

This is a generalised version of Maxwell's equation (M1): $\nabla \vec{E} = \rho / \epsilon_0$

In the integral form (lectures 1,2): $\int_S \vec{D} d\vec{S} = \int_V \rho_f dV$

Using the definition of electric susceptibility, $\vec{P} = \chi_E \cdot \epsilon_0 \cdot \vec{E}$,

$$\vec{D} = (1 + \chi_E) \cdot \epsilon_0 \vec{E} = \epsilon \epsilon_0 \vec{E}$$

where $\epsilon = 1 + \chi_E$ is the *relative permittivity* (aka the *dielectric constant*) of the medium.

Gauss law for uniform ϵ : $\nabla \vec{E} = \rho_f / (\epsilon_0 \epsilon)$

- ❖ The **E**-field *decreases* by a factor of $\epsilon > 1$ relative to the field produced by the same distribution of free charges in free space.
- ❖ Ideal conductors: $\vec{E} = 0$ (lecture 3), therefore $\epsilon \rightarrow \infty$.

Physical meaning

Assumptions behind the idea of a constant ϵ for a dielectric:

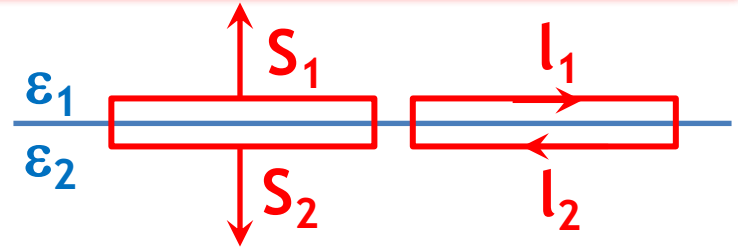
- ❖ **Linearity**: $\mathbf{P}$ is proportional to $\mathbf{E}$.
This breaks down at high electric fields.
 - ❖ **Isotropy**: polarisation vector $\mathbf{P}$ is collinear to $\mathbf{E}$.
NB: some crystals are anisotropic, particularly under mechanical stress (the piezoelectric effect).
 - ❖ **Homogeneity**: no dependence on the position within material.
- Most insulators satisfy these conditions in practice (and are known as “**LIH dielectrics**”).

Physical meaning:

- ❖ Electric field $\mathbf{E}$ [V/m]: *force* acting on unit charge.
- ❖ Polarisation $\mathbf{P}$ [C/m²]: induced *dipole moment* per unit volume, a local property of matter.
- ❖ Electric displacement $\mathbf{D}$ [C/m²]: introduced to facilitate field calculations in the presence of dielectrics (polarisation charges do not enter the Gauss law for $\mathbf{D}$ -field).

Boundary conditions for E and D

Boundary of two dielectrics
with relative permittivities ϵ_1 and ϵ_2



Gauss law for a very thin “pillbox” cylinder (no free surface charges):

$$\int_S \vec{D} d\vec{S} = 0 \quad \text{therefore} \quad \vec{D}_1 \vec{S}_1 + \vec{D}_2 \vec{S}_2 = 0$$

Using $\vec{S}_1 = -\vec{S}_2$, we obtain $\vec{D}_1 \vec{S}_1 = \vec{D}_2 \vec{S}_1$ and $D_{1n} = D_{2n}$

More generally, the discontinuity of D_n equals to surface density of *free charges*: $D_{1n} - D_{2n} = \sigma$

A very thin rectangular loop: infinitely small area, $\int_S \vec{B} d\vec{S} = 0$.
Therefore, Faraday’s law

$$\oint_L \vec{E} d\vec{l} = \vec{E}_1 \vec{l}_1 + \vec{E}_2 \vec{l}_2 = 0, \quad \text{leads to} \quad E_{1t} = E_{2t}$$

Field lines at boundaries

$\underline{D}_n$ is continuous:

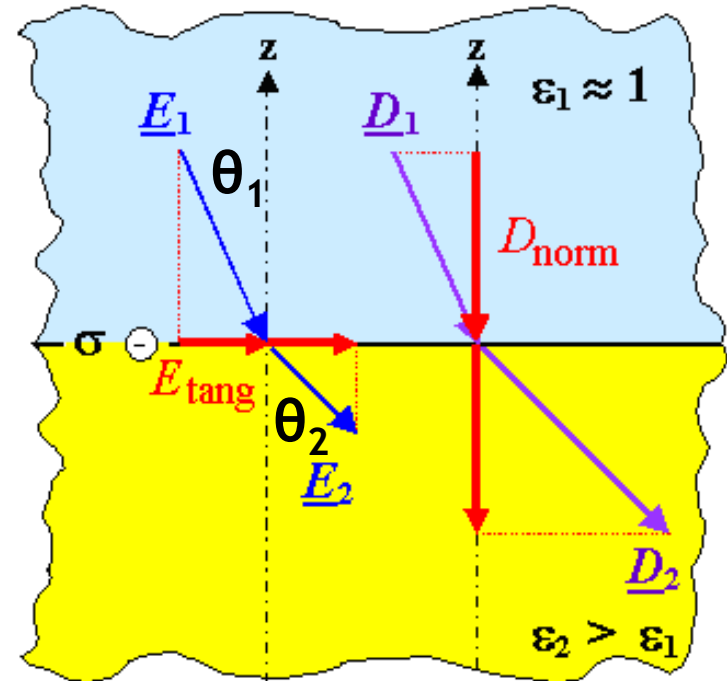
$$\epsilon_1 \underline{E}_1 \cos \theta_1 = \epsilon_2 \underline{E}_2 \cos \theta_2$$

$\underline{E}_t$ is continuous:

$$\underline{E}_1 \sin \theta_1 = \underline{E}_2 \sin \theta_2$$

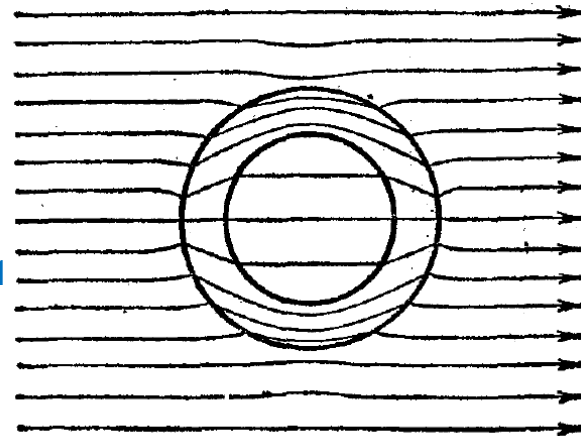
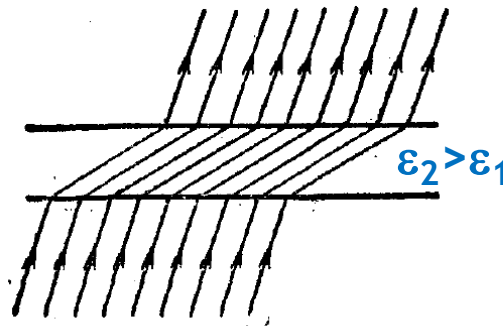
Therefore
$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\epsilon_1}{\epsilon_2}$$

(similar to Snell's law in optics – not by accident!)

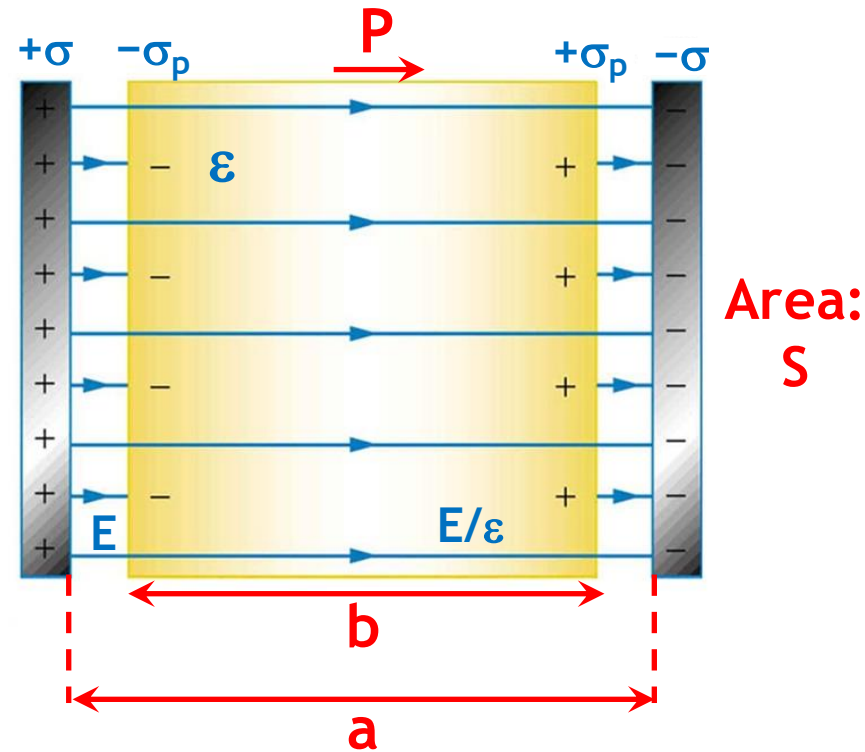
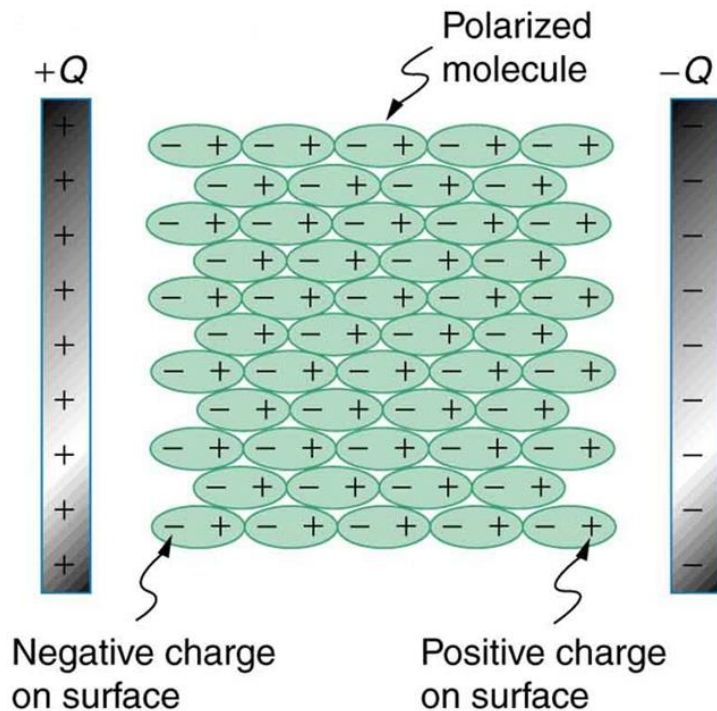


❖ Electrostatic field is partially shielded in cavities in high- ϵ dielectrics.

❖ Cf. complete shielding in a hollow conductor (Faraday's cage, *lecture 3*)



A dielectric slab in a capacitor (1)



- ❖ The displacement field $\mathbf{D}$ is uniform (because $\mathbf{D}_n$ is continuous).
- ❖ From Gauss law, $\mathbf{D} = \sigma$ everywhere inside the capacitor.
- ❖ Therefore $\mathbf{E} = \mathbf{D}/(\epsilon\epsilon_0)$ is *reduced* by a factor of ϵ in the dielectric.
- ❖ From Gauss law, surface polarisation charge: $\sigma_p = \sigma(\epsilon-1)/\epsilon$.

A dielectric slab in a capacitor (2)

Potential difference between the two plates:

$$V = \int_0^a E dx = \int_0^a \frac{D}{\epsilon_0 \epsilon} dx = \frac{D}{\epsilon_0} (a - b) + \frac{D}{\epsilon_0 \epsilon} b = \frac{\sigma}{\epsilon_0} (a - b) + \frac{\sigma}{\epsilon_0 \epsilon} b$$

For **b=0** (no dielectric slab), $V_1 = \frac{\sigma a}{\epsilon_0}$

For **b=a** (slab occupying all space), $V_2 = \frac{\sigma a}{\epsilon_0 \epsilon}$: *reduces* by a factor ϵ .

Capacitance in the two cases:

$$C_1 = \frac{Q}{V_1} = \frac{\sigma S}{V_1} = \frac{\epsilon_0 S}{a}$$

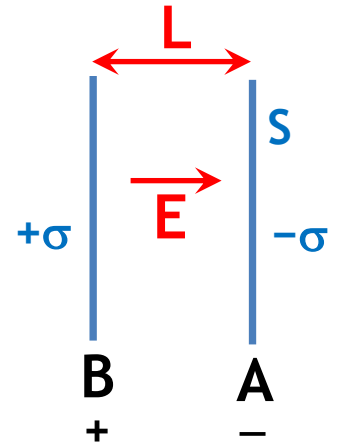
$$C_2 = \frac{Q}{V_2} = \frac{\sigma S}{V_2} = \frac{\epsilon_0 \epsilon S}{a} : \textit{increases} \text{ by a factor } \epsilon.$$

Energy density of the E-field (1)

Work required to move a charge $\delta q = S\delta\sigma$ from plane A to plane B in a parallel-plate capacitor filled with a dielectric material:

$$\delta W = EL \cdot \delta q = ELS \cdot \delta\sigma = VE \cdot \delta\sigma$$

(S : area; σ : surface charge density; V : volume)



Inside the capacitor, $D = \sigma$, therefore $\delta D = \delta\sigma$

Finally, $\delta W = VE \cdot \delta D$

The increment in *energy density* [J/m³]: $\delta w = \frac{\delta W}{V} = E\delta D$

More generally (also for anisotropic dielectrics), $\delta w = \vec{E}\delta\vec{D}$

Energy density of the E-field (2)

In the assumption $\vec{D} = \epsilon\epsilon_0\vec{E}$ (i.e. for LH dielectrics), the *energy density* is

$$w = \int_0^D dw = \int_0^D E dD = \epsilon_0\epsilon \int_0^E E dE = \frac{1}{2}\epsilon_0\epsilon E^2 = \frac{1}{2}ED$$

of which $\frac{1}{2}\epsilon_0(\epsilon - 1)E^2$ is due to polarisation of the dielectric.

The polarisation contribution often dominates (e.g. water: $\epsilon=81$).

More generally (also for anisotropic dielectrics), $w = \frac{1}{2}\vec{E}\vec{D}$

In electrostatics, field energy can be interpreted as the potential energy of electric charges (assuming action at a distance).

Electrodynamic interpretation: *energy is localised in space*, and is carried by the electric field.

Summary

- ❖ The electric displacement field is defined as

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

- ❖ Gauss law for the electric displacement field:

$$\nabla \cdot \vec{D} = \rho_f \quad \text{or equivalently} \quad \int_S \vec{D} d\vec{S} = \int_V \rho_f dV$$

- ❖ For linear, isotropic and homogeneous (e.g. most) dielectrics,

$$\vec{D} = (1 + \chi_E) \cdot \epsilon_0 \vec{E} = \epsilon \epsilon_0 \vec{E} \quad (\epsilon > 1)$$

- ❖ Boundary conditions (no free surface charges):

$$D_{1n} = D_{2n} \quad \text{and} \quad E_{1t} = E_{2t}, \quad \text{leading to} \quad \frac{\tan \theta_1}{\tan \theta_2} = \frac{\epsilon_1}{\epsilon_2}$$

- ❖ Energy density of the electric field: $w = \frac{1}{2} \vec{E} \cdot \vec{D}$